

Table 4 Ratio of total heat transfer area to planform base area for the enhanced surfaces of the present study

Enhanced surface Geometry	Total heat transfer surface area ^(a) A_t (cm ²)	Area ratio ^(b) $A_r^+ = \frac{A_t}{A_{chip}}$
Flat (reference)	1.61	1.00
Low profile microgroove (reference)	3.54	2.26
Low profile microstud (reference)	3.85	2.39
Short smooth stud	8.00	4.96
Long smooth stud	17.57	10.89
4 pin	9.92	6.15
9 pin	9.06	5.62
Axial microgroove stud	11.21	6.95
Radial microgroove stud	12.34	7.65
Microstud stud	14.74	9.14

^(a)Planform area of chip $A_{chip} = 1.61$ cm².

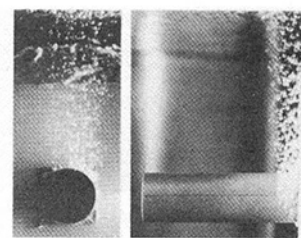
^(b) A_r included the heat transfer area of the extended surface plus the unfinned chip base area.

16 mm "short" smooth stud conformed to typical packaging limitations, while the 40 mm long smooth stud was tested for photographic analysis of the interactions of multiple modes of heat transfer on a horizontal cylinder.

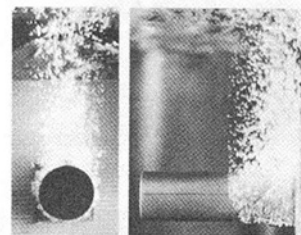
The 4-pin and 9-pin surfaces, Fig. 6, provided information on the performances of pin fin arrays. The spacing between the pins was set at one bubble departure diameter and the length-to-diameter aspect ratio was set at 2 to enforce practical spatial packaging requirements of multichip modules. The 4 and 9-pin structures were fabricated from a single bar of oxygen-free copper by electric-discharge machining (EDM) to alleviate the contact resistance that accompanies soldered assembly. Table 4 details the actual heat transfer areas and the ratios of total surface to chip base areas for each of the enhanced chip attachments.

The enhancement of boiling by extended surfaces can be measured in different forms as discussed in the previous section. To a packaging engineer charged with dissipating the maximum possible heat rate from an electronic device of fixed dimensions while maintaining the desired device temperature, the most meaningful measure of heat flux is total heat transfer rate divided by the surface area of the device; heat flux based on the total wetted area of the enhanced surface attachment is of little importance in this situation provided the enhancement meets the volumetric constraints in the packaging of multi-module electronic hardware. For this reason, unless specified otherwise, the boiling curves presented hereafter have been based on heat dissipation rate per unit chip planform area, 12.7×12.7 mm², and mean temperature at the base of the enhanced surface attachment, T_w .

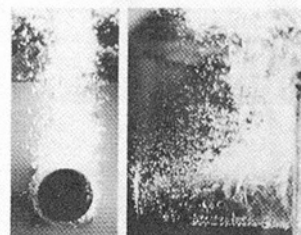
Flow Visualization of Boiling on Pin-Fin Surfaces. Several experiments were performed with the long, smooth stud to photograph the evolution of boiling pattern with increased heat flux. Figure 7 shows side and end views of the stud in saturated FC-87. One distinct feature of boiling behavior at heat fluxes slightly above incipience is the concentration of active nucleation sites near the upper side of the stud. This observation illustrates one weakness with axisymmetric numerical models for boiling on cylindrical fins. As heat flux was increased, the nucleation front advanced toward the tip with more circumferential boiling uniformity except for a narrow region close to the nucleation front. Close-up end-view examination of the surface showed the bubbles emanating normal to the stud surface with a uniform bubble departure diameter independent of circumferential location. However, this phenomenon was visually obscured by the buoyancy forces pulling the bubbles toward the upper surface of the stud. As heat flux was increased to only 27 percent of CHF, nucleation covered the entire perimeter except for the tip, which was partially undergoing natural convection in liquid. Boiling near the base of the stud



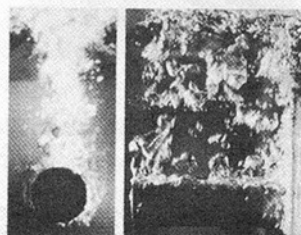
$q = 2.0$ W/cm²



$q = 5.6$ W/cm²



$q = 28.6$ W/cm²



$q = 105.8$ W/cm² (Following CHF)

Fig. 7 Development of boiling pattern on the long smooth stud in saturated FC-87 with increasing heat flux

was very vigorous and bubble coalescence into large vapor jets and columns was very noticeable. Isolated nucleate boiling was confined to regions near to or on the tip. Transition to film boiling occurred at $q_m = 105.8$ W/cm². The stud tip and perimeter were fully encased in a fairly smooth vapor layer which accumulated vapor at the upper surface of stud. The vapor was released from the crests of five vapor waves having a mean wavelength of 8.0 mm compared to a predicted Taylor's "most dangerous wavelength" in FC-87 of $\lambda_D = 7.87$ mm.

Figure 8 shows side and end views of boiling on the 4-pin surface in saturated FC-87. As in the case of boiling from a single cylindrical stud, nucleation was initiated near the base of the pins. At fluxes slightly above incipience, the nucleation front moved toward the pin tips, totally engulfing the upper pins, while the lower pins were partially undergoing natural convection. However, a slight increase in heat flux to approximately 6 percent of CHF activated nucleation over the entire surface. This would indicate that numerical models of boiling heat transfer from a pin fin array based on identical heat transfer performances of individual pins would be less successful for heat fluxes close to incipience. When the heat flux was increased to the CHF value of 89.8 W/cm², a single continuous vapor blanket engulfed the enhanced surface, al-